

DESIGN & ARCHITECTURE · IT & DIGITAL SERVICES

GEOTAG

FIELD INSPECTION MAPPER

GeoTag · Field Inspection Mapper

Design & Architecture Document

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How to read this document. Sections 1–10 are the **design** view (purpose, users, journeys, screens, visual system, accessibility) written for a general stakeholder audience. Sections 11–17 are the **architecture** view (system, data, hosting, security) for a technical/IT audience. Sections 18–21 cover **known limits, decisions (ADRs), and roadmap**. A one-page reviewer summary is at the end (§21).

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Design

1. Executive overview

GeoTag is a lightweight, installable web application that lets field personnel turn a geotagged photograph into a **precise, shareable record of *where something is*** — so the next person who has to go there can find the exact spot — without uploading the photograph anywhere.

A user drops one or more photos onto the page. For each photo the app reads the embedded GPS metadata **in the browser**, plots the location on an interactive map, looks up a human-readable address, and produces a side-by-side **photo + map** image with a coordinate caption. That artefact can be copied, downloaded, or shared into a report, a work order, or a message so the recipient can navigate straight to the location — and it works just as well for an inspection report as for any other field hand-off.

The defining architectural characteristic is **privacy by design**: photographs and their coordinates are processed entirely on the user's device. The only network traffic is for map tiles and an optional address lookup — never the photo itself. The app is a single static HTML file with vendored libraries, so it has **no backend, no database, no user accounts, and no server-side data storage**. It runs fully offline once installed.

This document describes the design, the user experience, the technology choices, the data-flow and privacy posture relevant to an IT/security review, the deployment model, the key decisions taken (as ADRs), and — at the request of the business — **how the application was built collaboratively** with an AI pair-programmer.

2. Website purpose & the stakeholder problem

2.1 The problem it solves

Field personnel routinely photograph a piece of equipment, a defect, or a developing issue out in the plant. Almost every one of those photos already carries the **exact GPS location** of where it was taken — but that location data normally stays locked inside the image file and is never put to use.

So the equipment's physical location ends up being communicated to the next person in *words*: a written description, nearby landmarks, or a rough bearing such as “northeast corner of Plant 12.” Whoever later has to **return to that exact spot** — a maintenance technician sent to fix the leak, an NDE technician performing a follow-up inspection, or operations personnel asked to monitor the equipment across shifts —

then has to interpret that description and hunt for the location. That is slow, ambiguous, inconsistent between staff, and easy to get wrong.

GeoTag closes this gap. It reads the photograph's own GPS data **on the device** and turns it into a precise, shareable location: an interactive map, exact coordinates, and a side-by-side photo + map artefact that points the next person straight to where the photo was taken — with one-tap links to open the spot in a phone's map app for turn-by-turn navigation.

Although it grew out of inspection work, the tool is **not limited to inspection reports**. It serves any field workflow where one person needs to show another exactly where something is and that second person will eventually have to go find it — fixing it, re-inspecting it, monitoring it, or simply confirming a concern. And because the image is processed entirely on the device, it keeps potentially sensitive site imagery off external servers.

2.2 The stakeholder problem

STAKEHOLDER	THE PROBLEM THEY HAVE TODAY	HOW GEOTAG ADDRESSES IT
Field personnel capturing the issue (inspectors & others)	Can only convey a photographed item's location in words, landmarks, or a rough bearing — imprecise, slow, and inconsistent between staff.	Captures the photo's actual coordinates plus a map the recipient can navigate to.
Personnel who must return to the location (maintenance, NDE, operations)	Must interpret a vague written description and then search the plant to find the exact equipment again.	Receive exact coordinates + map + photo, and can open the spot directly in a map app.
IT & Security	Field tools that upload imagery to external/consumer cloud services create data-governance and breach exposure.	No upload, no backend, no accounts — minimal attack surface to assess.
Digital Services / delivery	Building and operating a bespoke app is costly (servers, auth, patching).	A static, serverless app with near-zero operational footprint.
Records / compliance	Location claims in reports are hard to verify and easy to mis-transcribe.	Coordinates are read from the photo and rendered into the artefact verbatim, with an explicit flag if the pin was manually adjusted.

2.3 Design goals

1. **Privacy first** — image data must never leave the device.
2. **Zero install friction** — works from a URL; installable without an app store.
3. **Works in the field** — fully functional offline after first load.
4. **Actionable, shareable output** — a clean photo + map + coordinates the next person can navigate to (drops straight into a report, work order, or message).
5. **No operational backend** — nothing to host, patch, or breach server-side.

3. Stakeholders & user groups

GeoTag has a small number of distinct user types. Each row states what that user needs to *accomplish* with the app.

USER GROUP	WHO THEY ARE	WHAT THEY NEED TO ACCOMPLISH	PRIMARY SURFACE
Field personnel (inspectors & others)	On-site staff capturing a photo of an issue or piece of equipment, often on a phone, sometimes offline.	Drop a just-taken photo, confirm the location is right (and nudge it if not), and produce a precise location record to hand off.	Mobile PWA, single-column.
Returning personnel (maintenance / NDE / operations)	Whoever must physically go back to the photographed location later — to fix, re-inspect, or monitor it.	Find the exact spot from the shared coordinates and map, and open it in a map app for navigation.	Receives the report image / coordinates; uses the “Open in maps” links.
Report assembler / back-office	Office staff packaging the location records into reports or work orders, usually on desktop.	Batch-process several photos, copy/download each report image into a document.	Desktop, two-column, multi-photo.
IT / security reviewer	Assesses the tool before rollout.	Understand data handling, egress, hosting, and risk.	This document.
Application owner / maintainer	Owns the product and future changes.	Extend features, update dependencies, redeploy.	Source repo + this document.

There are **no authenticated roles, no admin panel, and no per-user configuration** — every user sees the same single-purpose tool. This is deliberate (see ADR-001, ADR-006).

4. Main user journeys & top actions

4.1 Top three actions

The interface is optimised around three actions, in priority order:

1. **Add photo(s)** — drag-and-drop or browse.
2. **Confirm / correct the location** — read the coordinates, optionally drag the pin and set the heading, edit the caption.
3. **Build & take the report image** — copy, download, or share.

4.2 Primary journeys

Every journey ends in an artefact whose job is to get the **next person to the exact location** — so the journeys span both creating the record and acting on it.

Journey A — Single photo (mobile, in the field). Open app → drop one photo → app reads GPS and shows the map and readout → (optional) drag pin / set heading / edit caption → tap **Build report image** → **Copy**

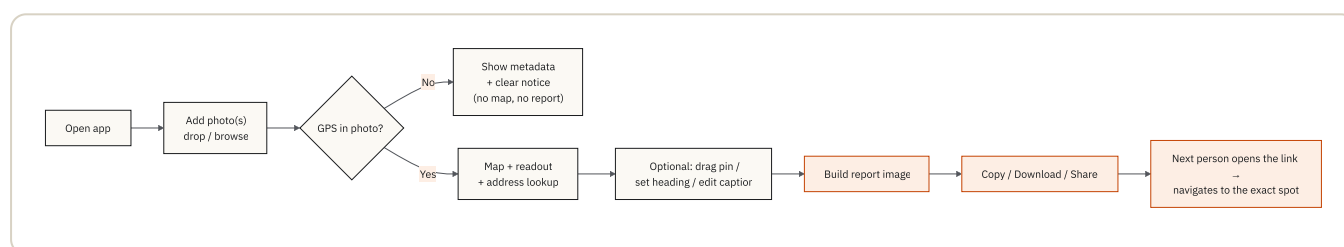
or **Share** into the report. Works offline for previously viewed map areas.

Journey B — Batch (desktop, back-office). Open app → drop several photos → each gets its own card, map, and report button → switch map style once (applies to all) → per card, adjust and **Build report image** → **Download PNG** for each → assemble into the report document.

Journey C — Correcting a bad fix. Photo has GPS drift → drag the pin to the true location → the readout updates live, the address re-looks-up, and the card/report is flagged “**manually adjusted**” so the correction is transparent in the evidence.

Journey D — Photo without GPS. Drop a photo with no GPS → app shows the available metadata (camera, capture time) and a clear notice, hides the map, and disables the report button — no dead-ends.

Journey E — Returning to the location (the recipient). A maintenance, NDE, or operations colleague receives the report image (or just the coordinates) → opens the “**Open in Google / Apple / OpenStreetMap**” link, or reads the coordinates → navigates straight to the exact spot where the photo was taken, instead of decoding a worded description. This is the payoff the other journeys feed.



5. Key features

FEATURE	DESCRIPTION
On-device EXIF/GPS read	Latitude/longitude (decimal & DMS), altitude, heading, capture time, and camera make/model parsed from the photo in the browser.
Multi-photo workspace	Any number of photos can be added; each gets its own card, map, and report button.
Interactive map	Leaflet map with Street (OpenStreetMap), Satellite (Esri World Imagery), and Topo (OpenTopoMap) styles.
Reverse geocoding	Optional human-readable address from OpenStreetMap Nominatim, auto-filled into an editable caption.
Draggable pin	The location pin can be nudged to correct GPS drift; the report is flagged as “ <i>manually adjusted</i> ” when this happens.
Heading / direction cone	Camera-pointing direction shown as a cone on the map; pre-filled from EXIF or set by hand.
HEIC support	iPhone <code>.heic</code> photos are converted to a viewable JPEG in-browser.
Report image export	Composited photo + map PNG with coordinate caption and product footer, rendered on a <code><canvas></code> .
Copy / Download / Share	Output can be copied to clipboard, downloaded as PNG, or shared via the native OS share sheet.
Installable PWA	Adds to home screen / desktop; launches in its own window.
Offline-capable	App shell, libraries, and fonts are precached; previously viewed map tiles are cached too.

6. Page / view-by-view design

GeoTag is a **single-page application** — one URL, one screen, no navigation between pages. What changes is which *views* are visible. Each view below lists its purpose and the **information hierarchy** (what the user should see first, second, and last).

6.1 Masthead / header (always visible)

- **Purpose:** identity and the privacy promise.
- **First:** the GeoTag wordmark and “Field Inspection Mapper” descriptor.
- **Second:** the `100% LOCAL` privacy tag.
- **Last:** the one-line explanation that photos are read in-browser and never uploaded.

6.2 Dropzone / intake (initial state)

- **Purpose:** the single call-to-action — get a photo in.

- **First:** the large drop target with the  icon and “Drop geotagged photos here”.
- **Second:** the accepted formats line (JPG · PNG · HEIC · add as many as you like).
- **Last:** the **Browse files** button for users who prefer a file picker.

6.3 Workspace toolbar (after first photo)

- **Purpose:** controls that apply across all photos.
- **First:** the **map-style** segmented control (Street / Satellite / Topo).
- **Second:** the **photo count**.
- **Last:** **Clear all**.

6.4 Per-photo card (the core working view)

One card per photo; the heart of the app.

- **First:** the card header — photo number, filename, and a remove (✕) control.
- **Second:** the **photo** (left/top) and the **map** with the location pin (right/bottom) side by side, so the user immediately sees “*this photo* → *this place*”.
- **Then:** the **readout** — coordinates (decimal & DMS), altitude, editable **heading**, capture time, camera, the editable **caption**, and “Open in” map links.
- **Last:** the **Build report image** button (the card’s primary action) and a per-card status line.

6.5 Export modal (on Build)

- **Purpose:** preview and take the finished artefact.
- **First:** the rendered **report image** preview.
- **Second:** the take-away actions — **Copy image**, **Share** (when supported), **Download PNG**.
- **Last:** the close control. Dismissible by ✕, backdrop click, or .

6.6 Ambient feedback — status banner & toast

- **Status banner** (under the dropzone, and per card): inline progress and errors, colour-coded (ok / warn / error) with a spinner while working.
- **Toast:** brief confirmations (“Copied ✓”, “Downloaded ✓”) that auto-dismiss.

7. Visual design system

7.1 Visual style & the message it sends

The aesthetic is a **field instrument / surveyor’s document**: a warm paper background with a faint blueprint grid, precise monospaced labels, framed corner ticks on the masthead, and a single hi-vis safety-orange accent. The intended message is “**a precise, trustworthy professional tool**” — closer to a calibrated instrument or an official form than a consumer photo app — which reinforces the credibility of the evidence it produces.

7.2 Colour palette (CSS custom properties, `index.html` :root)

TOKEN	VALUE	USE
<code>--paper</code>	<code>#ece7dc</code>	App background (with blueprint grid)
<code>--card</code>	<code>#fbf9f4</code>	Card / panel surfaces
<code>--ink</code>	<code>#1a1c1e</code>	Primary text & strong borders
<code>--ink-soft</code> / <code>--ink-faint</code>	<code>#52555b</code> / <code>#8b8e94</code>	Secondary / tertiary text
<code>--line</code> / <code>--line-strong</code>	<code>#d8d2c4</code> / <code>#c2bba9</code>	Dividers & borders
<code>--accent</code> / <code>--accent-deep</code>	<code>#ef5d12</code> / <code>#c54405</code>	Hi-vis surveyor orange (primary action, pin)
<code>--good</code>	<code>#15706b</code>	Success states

Colour is used sparingly: orange marks *action and location*; teal marks *success*; red marks *errors*. The map pin and direction cone reuse the same orange so the brand and the data point are visually unified.

7.3 Typography

- **IBM Plex Sans** — body and controls.
- **IBM Plex Sans Condensed** — the wordmark and section headings (uppercase, tracked-out) for an “official document” feel.
- **IBM Plex Mono** — all data values, field labels, coordinates, and captions, so numbers align and read as instrument output. Fonts are self-hosted woff2 (no third-party font CDN).

7.4 Spacing, shape & elevation

Consistent radius (`--radius: 14px`) on cards/panels, a shared soft shadow token, and generous padding. A 30px blueprint grid underlies the page. Layout is a centred shell capped at ~1080px on desktop.

7.5 Components & repeating interface patterns

The same handful of patterns repeat everywhere, which keeps the app learnable:

- **Cards** — every photo is a bordered card with a numbered header.
- **Panes with mono headers** — the photo and map each sit in a pane with a small uppercase mono label and a status dot.
- **Segmented control** — the map-style switch.
- **Buttons** — three weights: solid dark (default), **accent orange** (primary: Build / Download), and ghost (secondary: Clear all / Share).
- **Copy buttons** — a small icon button beside any copyable value (coordinates).
- **Status pills** — colour-coded inline banners with an optional spinner.
- **Toast** — transient confirmation.
- **Form fields** — caption textarea and numeric heading input share the same bordered, mono, orange-focus styling.
- **Icons** — line-style SVG (the crosshair glyph, copy, pin), matching the instrument theme.

8. Usability & design decisions

8.1 Choices that reduce confusion

- **One primary action at a time** — a big single dropzone to start; per card, a single accent **Build report image** button.
- **Immediate spatial proof** — photo and map are shown together so the user never has to imagine the link.
- **Plain-language privacy** stated up front, removing the “is my photo being uploaded?” doubt.
- **Honest state** — a manually moved pin is labelled *adjusted* on screen and in the report rather than silently overriding the EXIF location.
- **No dead-ends** — missing GPS, wrong file types, and offline lookups all return a clear message and a still-usable screen.

8.2 Choices that make it easier on mobile

- Single-column layout below 760px; the map fixes to a usable 300px height.
- The action bar becomes **sticky at the bottom** with full-width buttons, so *Build* is always in thumb reach.
- Native **Share** is offered when the device supports it.
- `viewport-fit=cover` and safe-area insets respect notches/home indicators.
- Installable to the home screen; launches full-screen like a native app.

8.3 Choices that make it easier on desktop

- Two-column **photo + map** side-by-side; the map height tracks the photo height (via a `ResizeObserver`) so the pair stays balanced.
- Wider shell (up to 1080px) and a right-aligned action row.
- Keyboard support (drop target is focusable and `Enter` / `Space` activates it; `Esc` closes the modal).
- Hover affordances on buttons and copy controls.

8.4 What is intentionally simple, and why

- **No accounts, settings, or navigation** — the tool does one job; configuration would add cognitive load and a data-storage surface for no benefit (ADR-006).
 - **No framework or build step** — keeps the app a single auditable file and the smallest possible attack surface (ADR-002).
 - **One artefact type (PNG)** — universally paste-able into any report tool; richer formats are deferred (see §20).
-

9. Interaction states & error handling

9.1 What each action triggers

USER ACTION	FEEDBACK / STATE
Drop / select photo	Upload + loading — “Reading image...” with spinner; HEIC shows “Converting HEIC...”.
GPS found	Success — “GPS location found”; map and readout render.
Address lookup running	Caption placeholder shows “Looking up address...”; fills when ready.
Build report image	Loading — button shows “Rendering...”; then the export modal (a confirmation/preview) opens.
Copy image	Confirmation toast — “Image copied ✓”.
Download PNG	Download triggered + “Downloaded ✓” toast.
Share	Hands off to the native OS share sheet (only shown when supported).
Remove card / Clear all	Card(s) removed; workspace collapses when empty.

There are **no “saved results”** to manage — nothing persists between sessions by design (see §13.2).

9.2 When the user makes a mistake

- **Wrong file type:** non-image files are filtered; if none are valid, an error banner says “Please choose JPG, PNG or HEIC files”; a partial batch shows “Some files were skipped”.
- **Pin dragged wrongly:** simply drag again; the *adjusted* flag and live readout keep it honest; the original EXIF location is retained internally.
- **Closed the modal by accident:** the card and its **Build report image** button remain; nothing is lost.

9.3 When data is missing, invalid, or incomplete

- **No GPS in the photo:** warning banner “No GPS data in this photo — showing available metadata”; the map pane is removed and the report button disabled, but camera/time metadata still shows.
- **Unreadable/corrupt image:** “Could not process this image. Try another file.”
- **Address lookup fails or offline:** the caption falls back to a typed-caption prompt (“Address lookup unavailable (offline) — type a caption”); coordinates and map are unaffected.
- **Map capture fails during export:** the report still renders with a “map unavailable” placeholder rather than failing the whole image.
- **Clipboard/Share unsupported:** copy falls back to “Copy not supported — use Download”.

10. Accessibility & device support

10.1 Target standard

The app targets **WCAG 2.1 Level AA** as the design baseline. Current implementation includes several AA-supporting measures (below); a formal conformance audit is listed as outstanding work (§18, §20).

10.2 What is implemented today

- **Semantic roles & labels:** the dropzone uses `role="button"`, `aria-label="Upload photos"`; the map-style switch uses `role="radiogroup"` with an `aria-label`; the workspace is an `aria-live="polite"` region so status changes are announced.
- **Labelled controls:** action buttons carry `aria-label` / `title` (e.g. “Remove photo”, “Copy”, “Close”); the photo `<img>` has descriptive `alt` text.
- **High contrast:** dark ink on warm paper; the orange accent is used with dark borders/text rather than as the sole signal.
- **Focus styles:** inputs show an orange focus border; controls are reachable.

10.3 Keyboard-only use

- The drop target is focusable (`tabindex="0"`) and activates on **Enter/Space**.
- **Browse files**, buttons, links, and form fields are standard focusable elements.
- The export modal closes on **Esc** (and backdrop click).
- *Known gap:* the modal does not yet trap focus, and the Leaflet map’s full keyboard panning is not customised — see §18.

10.4 Screen-reader labelling of images, icons, maps, buttons, fields

- Photos have `alt` text; decorative icons are marked `aria-hidden`.
- Buttons and the copy controls have text or `aria-label`s.
- *Known gap:* the **map** is a visual control; its pin/coordinate state is available as text in the readout, but the interactive map itself is not fully described to assistive tech. The coordinate readout is the accessible equivalent.

10.5 Smallest expected phone size

Layout is verified to remain usable down to a **~320px-wide** viewport (single column, sticky full-width action bar, 300px map). Safe-area insets handle notched devices.

10.6 Browsers & devices supported

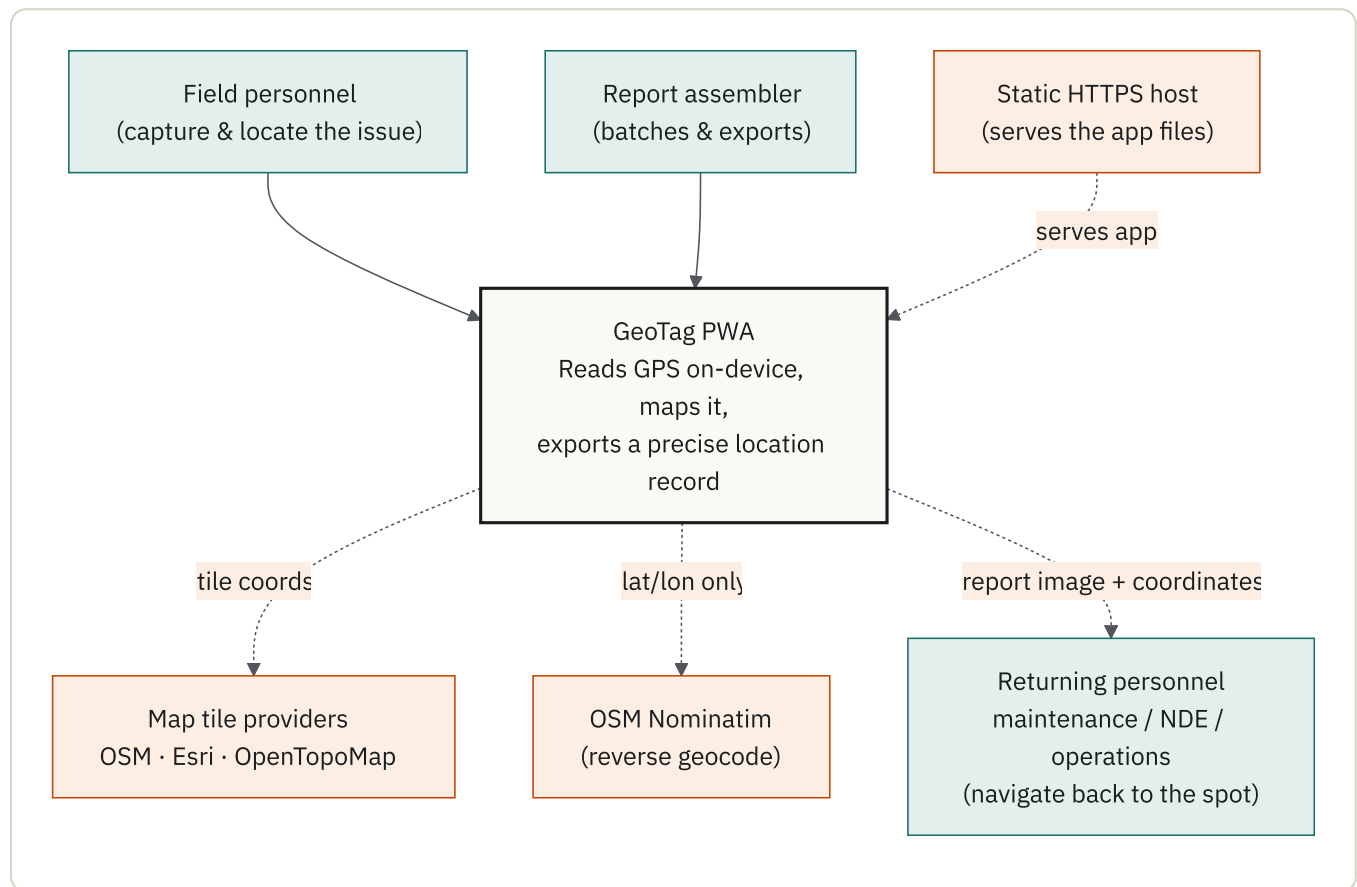
Current evergreen browsers on desktop and mobile (Chrome, Edge, Safari, Firefox). Uses widely-supported platform features: File API, Canvas, Service Workers, Clipboard and Web Share where available (Share is feature-detected and hidden when unsupported). Installable on iOS/iPadOS (Safari → *Add to Home Screen*), Android (Chrome → *Install app*), and desktop Chrome/Edge.

Architecture

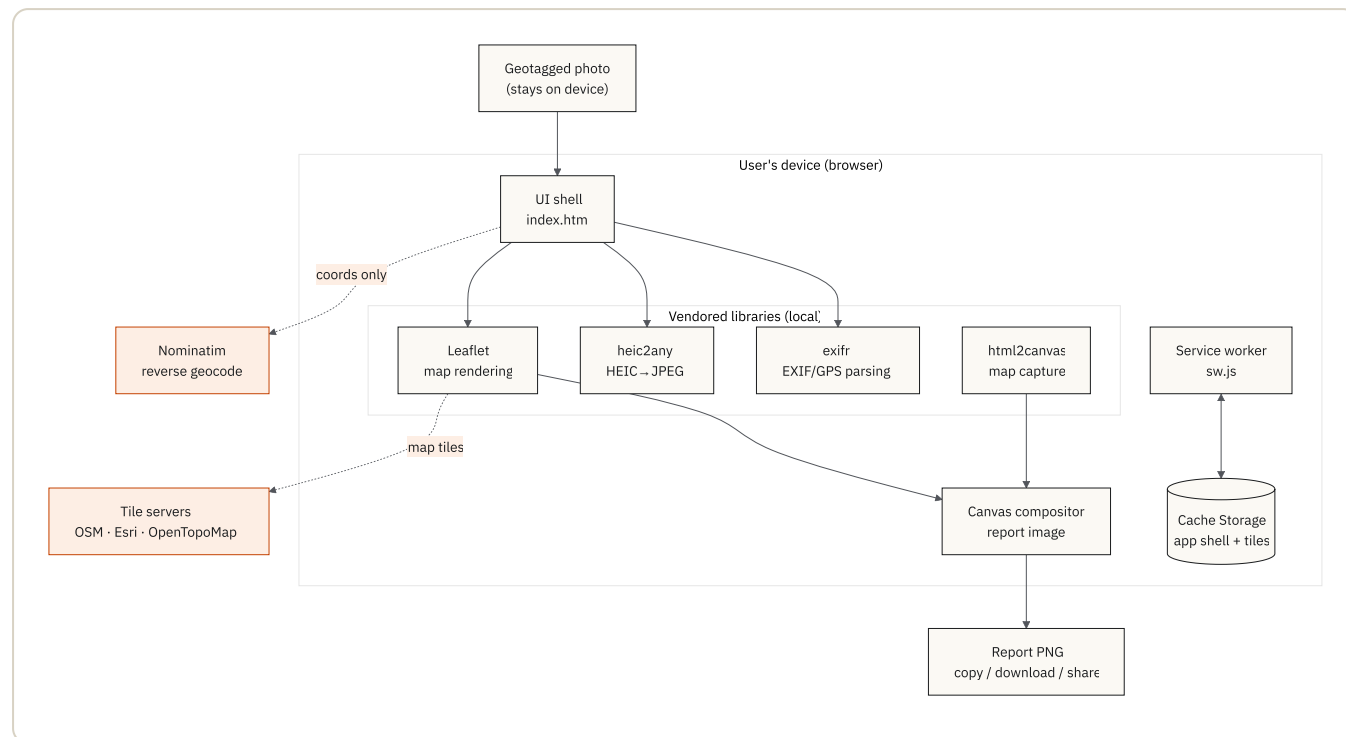
11. System architecture overview

GeoTag is a **single-page, client-side static application**. There is no application server — the “backend” is simply a static file host plus a small number of third-party tile/geocoding services called directly from the browser.

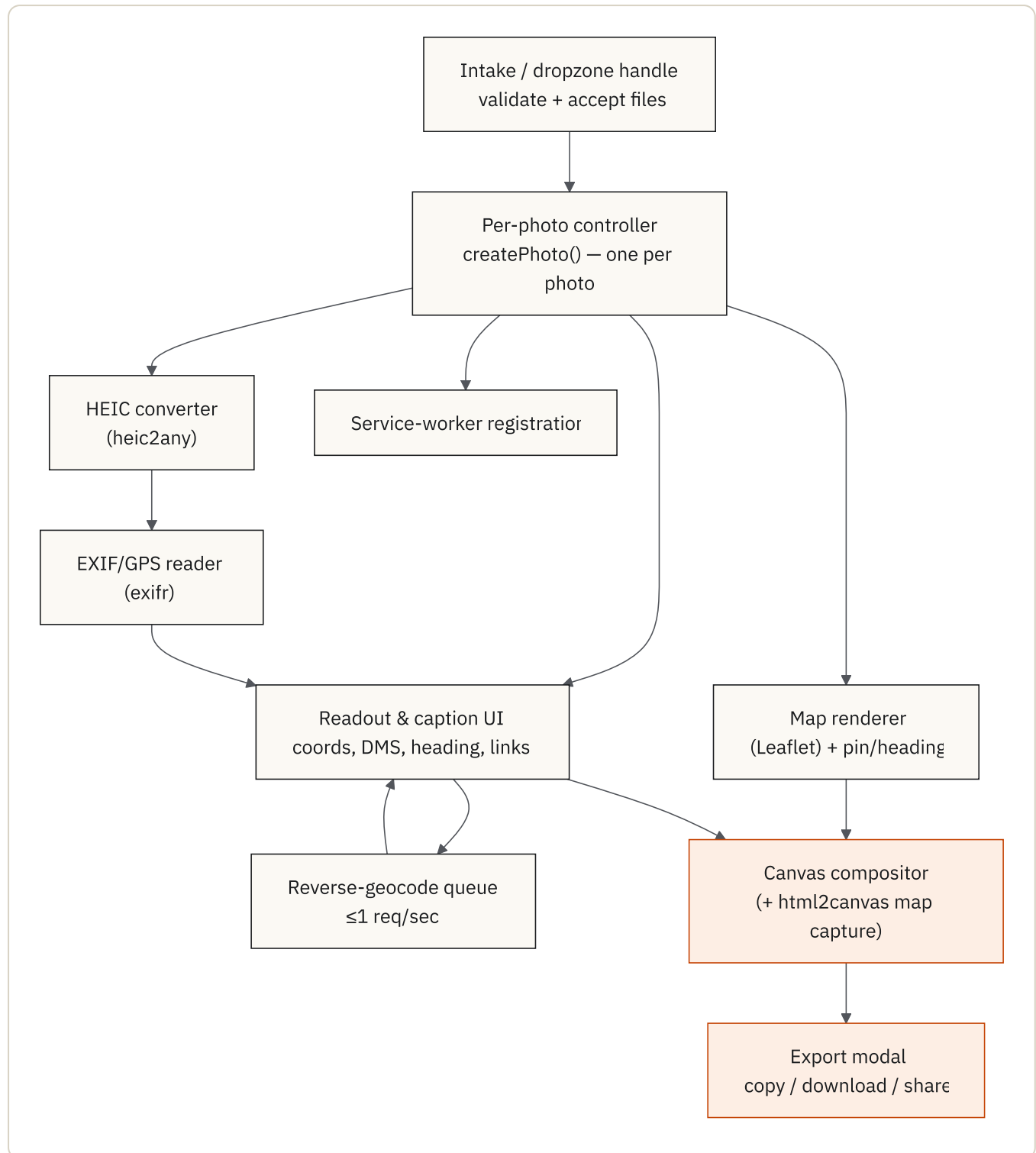
11.1 System context (C4 — level 1)



11.2 Containers (C4 — level 2)



11.3 Components (C4 — level 3, inside [index.html](#))



11.4 Logical layers

1. **Presentation / shell** — [index.html](#) contains the markup, an inline CSS design system, and the application logic in a single IIFE.
2. **Processing libraries** — four vendored libraries (see §12) handle EXIF parsing, HEIC decoding, map rendering, and DOM-to-canvas capture.

- 3. **Compositor** — a `<canvas>` routine draws the photo, a clean off-screen map, the pin/heading cone, and the caption into the final report image.
- 4. **Offline layer** — `sw.js` precaches the app shell and runtime-caches tiles.

11.5 Per-photo component model

Each uploaded photo is handled by an isolated `createPhoto()` instance owning its own card DOM, Leaflet map, marker, and state object (`{lat, lon, heading, caption, pinMoved, ...}`). Instances are tracked in a single `photos[]` array. Global controls — map-style switch, *Clear all*, window resize — fan out to every instance. This keeps multi-photo behaviour predictable without a framework or shared mutable global state.

12. Technology stack

GeoTag deliberately uses **no build step and no framework**. It is plain HTML/CSS/JavaScript, served as static files. All third-party code is **vendored** (committed into `vendor/`) rather than loaded from a CDN, which guarantees offline operation and removes runtime dependency on external script hosts.

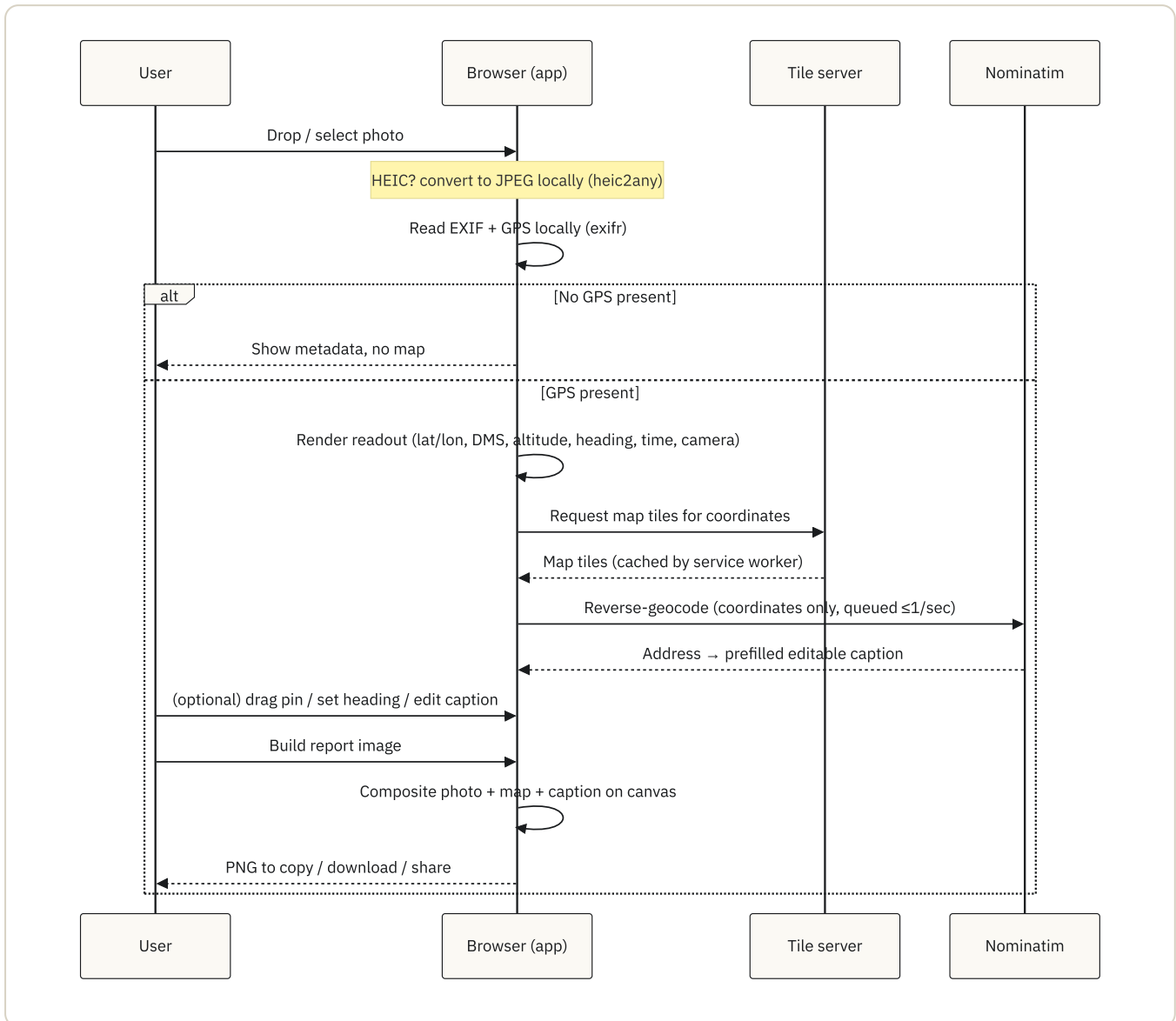
COMPONENT	VERSION	PURPOSE	LICENCE
Leaflet	1.9.4	Interactive map rendering	BSD-2-Clause
exifr	(vendored UMD)	EXIF / GPS metadata parsing	MIT
html2canvas	1.4.1	Capture the live map into the export canvas	MIT
heic2any	(vendored)	Convert HEIC photos to JPEG in-browser	MIT
IBM Plex fonts	—	Sans / Sans Condensed / Mono (self-hosted woff2)	SIL OFL 1.1

Note for IT: all of the above are permissively licensed (MIT / BSD / OFL) and are pinned and vendored. There is no `npm install`, no transitive dependency tree resolved at build time, and no package registry in the runtime path.

Why no framework / no build: the app is small enough that a framework would add more weight than it saves; a single static file is trivially auditable, trivially hosted, and has the smallest possible attack surface; and there is no build-time supply-chain step to secure. (Recorded as ADR-002.)

13. Data flow, storage & privacy

13.1 Lifecycle of a single photo



13.2 What data is collected, where it lives, and for how long

DATA	COLLECTED?	WHERE IT IS PROCESSED	STORAGE / RETENTION	LEAVES DEVICE?
Photograph (pixels)	Read locally, not collected by any server	In-browser only	In memory for the session; object URLs revoked on removal	No
EXIF / GPS metadata	Read locally	In-browser only	In-session memory only	No (except coordinates, below)
Coordinates (lat/lon)	—	In-browser	In-session memory only	Only to the chosen tile server and Nominatim, to fetch a map and an address
Address (geocode result)	Received	In-browser	In-session memory only	No (received)
Map tiles	Cached	Service worker	Cache Storage , capped at 300 tiles, oldest trimmed	Imagery received from tile host
App shell / libraries / fonts	Cached	Service worker	Cache Storage until cache version bumped	No
Final report image	Generated	In-browser	Not stored — exists until the user copies/downloads/shares	No, until the user chooses to

Nothing user-entered persists between sessions. There are no cookies, no `localStorage` / `IndexedDB` of user content, no accounts, and no analytics. The only browser storage used is the service-worker **Cache Storage** (static assets + map tiles), which holds no personal data beyond imagery of areas the user viewed.

13.3 What the website can create (outputs)

A single artefact type: a **PNG report image** per photo (photo + map + coordinate caption + product footer), produced on-device and taken away via copy / download / share. No files are written server-side.

13.4 What still requires an internet connection

- **Fresh map tiles** for areas not already cached (previously viewed areas work offline).
- **Reverse geocoding** (address) — online only; absence degrades to a typed caption. Everything else — reading GPS, plotting cached areas, pin/heading editing, and building/exporting the report — works fully offline.

13.5 Network egress (the complete list)

HOST	PURPOSE	DATA SENT
<code>tile.openstreetmap.org</code>	Street map tiles	Tile coordinates (derived from location)
<code>server.arcgisonline.com</code>	Esri satellite tiles	Tile coordinates
<code>tile.opentopomap.org</code>	Topographic tiles	Tile coordinates
<code>nominatim.openstreetmap.org</code>	Reverse geocode (address)	Latitude/longitude only

There are **no other endpoints** and **no first-party backend**. If these hosts are blocked, the app still reads photos and shows coordinates; only map imagery and address lookup degrade gracefully.

14. Security, privacy & risk register

14.1 Client-side security characteristics

- **Safe rendering:** all user- and EXIF-derived strings are HTML-escaped before DOM insertion (`esc()`), mitigating XSS from malicious file metadata.
- **CORS-clean canvas:** tiles are requested with CORS and cached as-is so the export canvas is never “tainted”, keeping copy/download/share working.
- **No `eval`, no dynamic remote script:** all executable code is the vendored, reviewable libraries plus inline logic; nothing is fetched and executed at runtime.
- **Graceful failure:** missing libraries, missing GPS, failed conversions, and offline geocoding all produce a clear message rather than a broken state.

14.2 Environment, secrets & configuration

- **No environment variables, secrets, or API keys are required** for the default tile and geocoding providers. There is nothing to rotate or leak.
- Configuration that exists is **source-level constants**: the tile/geocode URLs in `index.html` and the cache version + tile hosts in `sw.js`.

14.3 Security risks

RISK	ASSESSMENT / MITIGATION
Third-party script trust (vendored libs)	Pinned, vendored, reviewable; update only deliberately. Supply-chain risk is at <i>update time</i> , not runtime.
Host compromise / asset tampering	Mitigated by HTTPS + recommended strict CSP (§16.5); all scripts are same-origin.
Malicious image metadata	Escaped on output; parsing is sandboxed to the browser.
Clickjacking	Mitigated by recommended <code>X-Frame-Options</code> / <code>frame-ancestors</code> headers.

14.4 Privacy risks

RISK	ASSESSMENT / MITIGATION
Coordinates exposed to third parties	Map tiles and Nominatim necessarily receive location (as tile coords / lat-lon) and the device IP. This is the <i>only</i> data that leaves the device. Mitigation: repoint to org-hosted/commercial providers (ADR-004/005, §20).
Cached map imagery on shared devices	Cache Storage retains viewed tiles; clearable via the browser / “Clear site data”. No personal content is cached.
Accidental over-share of the report	The report is created locally; the user controls when/where it goes.

14.5 Failure points

FAILURE POINT	EFFECT	BEHAVIOUR
Tile server unavailable	No fresh map imagery	Cached tiles still serve; export shows “map unavailable” placeholder if needed.
Nominatim unavailable	No auto-address	Caption falls back to manual entry.
<code>html2canvas</code> map capture fails	—	Report still renders with placeholder map.
HEIC conversion fails	That photo not displayed	Clear per-card error; other photos unaffected.
Clipboard/Share API absent	Can’t copy/share	Falls back to Download.
Service worker unsupported/blocked	No offline	App still works online.

15. Offline & Progressive Web App design

GeoTag is a fully installable PWA defined by `manifest.webmanifest` (name, icons, theme colour, standalone display) and powered by the `sw.js` service worker.

Caching strategy:

- **App shell — cache-first.** On install the service worker precaches everything needed to open with no network: `index.html`, all vendored libraries, fonts, and icons (`SHELL_ASSETS`). Navigations fall back to cached `index.html` offline.
- **Map tiles — stale-while-revalidate, capped.** Tiles serve from cache immediately and refresh in the background into a separate cache capped at **300 tiles** (oldest trimmed) so storage stays bounded.
- **Everything else (e.g. Nominatim) — pass-through,** uncached; degrades gracefully.

Versioning / updates: the cache name carries a version (`CACHE_VERSION = "geotag-v2"`). Bumping it on any asset change installs a fresh shell cache and deletes stale caches on activation, so clients pick up new builds cleanly.

16. Hosting, deployment & operations

16.1 Hosting model

GeoTag is a set of **static files** served from the repository root, hostable on essentially any static host: GitHub Pages, Netlify, Cloudflare Pages, Azure Static Web Apps, S3 + CloudFront, or an internal static web server / intranet share over HTTPS. There is **no runtime, no server process, no database, and no scheduled job**. Operationally: “deploy the files and serve them over HTTPS.” For a Suncor deployment this would be an approved internal static host rather than a public provider (see §20.1).

16.2 Source code & project layout

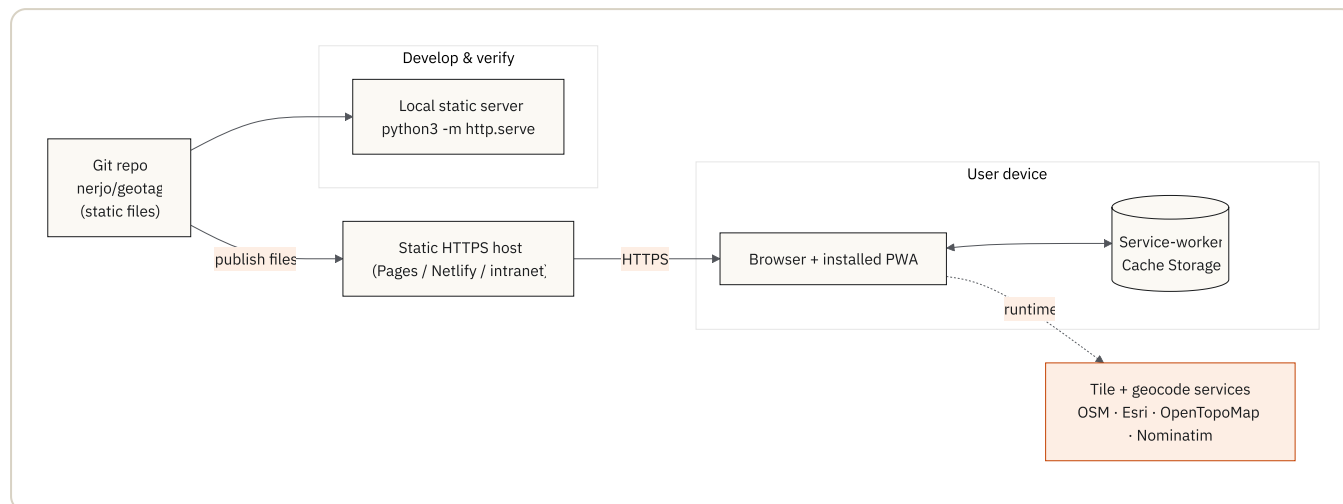
Source is stored in the Git repository `nerjo/geotag`.

<code>index.html</code>	App shell, design system, and all logic
<code>manifest.webmanifest</code>	PWA metadata (name, icons, theme)
<code>sw.js</code>	Service worker (offline shell + tile cache)
<code>vendor/</code>	Vendored, offline-capable libraries & fonts
<code> leaflet/</code>	
<code> exifr/</code>	
<code> html2canvas/</code>	
<code> heic2any/</code>	
<code> fonts/</code>	
<code>icons/</code>	App icons (192 / 512 / maskable / apple-touch / favicon)
<code>docs/</code>	This document (Markdown + PDF)
<code>README.md</code>	Quick start & install notes

16.3 Build, test & deploy

- **Build:** none — there is no compile/bundle step; the files are shipped as-is.
- **Test:** validation is currently **manual** — exercised with real geotagged, no-GPS, and HEIC photos across desktop and mobile, plus offline checks. There is **no automated test suite yet** (noted in §18/§20). The PDF of this document is generated from the Markdown via a headless-Chrome toolchain.
- **Deploy:** copy the repo contents to the static host over HTTPS; bump `CACHE_VERSION` in `sw.js` when any precached asset changes.

16.4 Deployment view



16.5 Recommended deployment hardening

- Serve over **HTTPS** (required for service workers and clipboard/share APIs).
- Apply a strict **Content-Security-Policy** (`script-src 'self'`, `connect-src` / `img-src` limited to the four hosts in §13.5 plus `data:` for the inline pin imagery). Because all scripts are same-origin, a strict CSP is straightforward.
- Standard static headers (HSTS, `X-Content-Type-Options: nosniff`, `Referrer-Policy`, `X-Frame-Options` / `frame-ancestors`).

16.6 Maintenance

- **Dependency updates** are manual and intentional: replace the file under `vendor/`, re-test, bump `CACHE_VERSION`. No automated dependency drift.
- **Small surface** — one HTML file plus four vendored libraries — keeps review cheap. **No secrets** are embedded.

17. How the application was built

At the business's request, this section documents *how* GeoTag was created, because the working method is itself notable: it was built as a **human-AI collaboration**, with the application owner directing the product and **Claude Code** (an agentic coding assistant) acting as a pair-programmer that wrote, refined, and explained the implementation.

17.1 The working method

Development was **conversational and iterative**, in short cycles:

1. **Describe the intent** in plain language (“read a photo’s GPS on-device and show it on a map”, “let me drag the pin to correct it”, “export a report image”).
2. **The AI implemented it** directly in the codebase and explained the trade-offs (e.g. why EXIF is read from the *original* file even after HEIC conversion, because conversion can strip metadata).

3. **Review on real behaviour** — the owner tried it with real photos; adjustments came in the next cycle.

17.2 How the product grew (from version history)

STEP	COMMIT	WHAT IT ADDED
1	<i>Initial commit</i>	Project scaffold.
2	<i>Base standalone app</i>	On-device GPS read, map, and export.
3	<i>Captions, draggable pin, heading control, report notes</i>	Editable caption (auto-filled from reverse geocoding), pin-drag correction with an “adjusted” flag, heading/direction control.
4	<i>Multiple photos, each with its own report button</i>	The multi-photo workspace and per-photo report generation.

Core capability first, then field-usability refinements, then scale to many photos — mirroring how the requirements were understood through use.

17.3 Why this matters for IT & Digital Services

- **Transparency** — the entire app is human-readable source in one repository; no opaque compiled artefact or hidden service.
- **Maintainability** — small, commented, framework-free code, extendable by a developer or by continuing the same AI-assisted workflow.
- **Auditability of method** — the incremental git history is a reviewable record of how each capability was introduced.

Decisions & outlook

18. Known limits & out of scope

18.1 Known limitations of the current version

- **No automated tests** — validation is manual (§16.3).
- **Accessibility gaps** — no focus-trap in the modal; the interactive map is not fully described to assistive tech (the coordinate readout is the accessible equivalent); no formal WCAG audit yet.
- **Third-party tile/geocode dependency** — default providers are public community services with their own usage policies and no SLA.
- **One output format** — PNG only; no combined/multi-photo or PDF export.
- **Single-pin reverse geocoding rate** — Nominatim is queried $\leq 1/\text{sec}$, so large batches fill addresses gradually.
- **No persistence** — closing the tab discards the working set (by design).

18.2 Explicitly out of scope (right now)

User accounts/auth, server-side storage or a database, analytics/telemetry, editing/annotating the photo itself, and integration into downstream asset/inspection systems.

18.3 Design choices that may need stakeholder approval

- **Use of public OSM/Esri/OpenTopoMap/Nominatim services** vs. an org-hosted or commercial provider (data-egress / usage-policy / SLA decision) — at Suncor, the SunMaps / SiteView option and its licensing/CORS feasibility (see §20.1).
- **Internal hosting location & approval** — an approved Suncor static-hosting location and HTTPS endpoint (see §20.1).
- **Branding** of the report image (logo/footer) for official use.
- **Accessibility conformance level** to be formally certified.
- **Retention/clearing policy** for cached map imagery on shared devices.

18.4 Future changes that would force architecture changes

Anything that breaks the “no backend” premise: saving/sharing reports server-side, multi-user collaboration or accounts, centralised audit logging, or server-side PDF generation would introduce a backend, storage, and authentication — a materially different architecture and security profile.

19. Architecture Decision Records (ADRs)

Each record captures the decision, context, options, rationale, trade-offs, risks, and status.

ADR-001 — Client-side only, no backend

- **Decision:** process everything in the browser; ship no server/API/database.
- **Context:** the core requirement is that photos never leave the device.
- **Options:** (a) client-only; (b) thin backend for geocoding/storage; © full SaaS with accounts.
- **Rationale:** (a) is the only option that structurally guarantees the privacy promise and removes server-side breach/operations cost.
- **Trade-offs:** no server-side persistence, audit, or cross-device sync.
- **Risks:** features needing shared state require re-architecture (§18.4).
- **Status:** Accepted (owner).

ADR-002 — No framework, no build step (single-file app)

- **Decision:** plain HTML/CSS/JS in one file; no bundler/framework.
- **Context:** small, single-purpose tool; auditability and minimal attack surface prioritised.
- **Options:** React/Vue + bundler; lightweight framework; vanilla single file.
- **Rationale:** the app is small enough that a framework adds more weight and supply-chain surface than it saves; one file is trivially auditable and hosted.

- **Trade-offs:** less structure for large future growth; manual DOM code.
- **Risks:** if scope grows substantially, maintainability could suffer.
- **Status:** Accepted (owner).

ADR-003 — Vendored libraries instead of CDN

- **Decision:** commit Leaflet/exifr/html2canvas/heic2any/fonts into `vendor/`.
- **Context:** must work fully offline and not depend on external script hosts.
- **Options:** CDN-loaded libraries; vendored libraries.
- **Rationale:** vendoring guarantees offline operation, pins versions, and removes runtime third-party script trust.
- **Trade-offs:** updates are manual; repo is larger.
- **Risks:** vendored copies can drift from upstream security fixes if not tracked.
- **Status:** Accepted (owner).

ADR-004 — Map providers: Leaflet + OSM / Esri / OpenTopoMap

- **Decision:** render with Leaflet; offer OSM, Esri World Imagery, OpenTopoMap tiles.
- **Context:** need free, key-less, multi-style maps that keep the export canvas CORS-clean.
- **Options:** commercial SDKs (Google/Mapbox, require keys/billing); open tiles + Leaflet.
- **Rationale:** open tiles need no API keys/secrets and cover street/satellite/ topo; Leaflet is small, permissively licensed, and CORS-friendly.
- **Trade-offs:** public tiles have usage policies and no SLA.
- **Risks:** availability/rate limits; coordinates + IP exposed to tile hosts.
- **Future work:** repoint to org-hosted/commercial tiles (small code change).
- **Status:** Accepted (owner); provider choice flagged for stakeholder approval.

ADR-005 — Reverse geocoding via OSM Nominatim (online-only, coordinates-only)

- **Decision:** look up addresses from Nominatim, sending only lat/lon, queued ≤ 1 /sec.
- **Context:** a human-readable caption is useful but must not weaken privacy.
- **Options:** no geocoding; Nominatim; commercial geocoder (keys/billing).
- **Rationale:** Nominatim is free and key-less; sending only coordinates keeps the privacy posture; rate-limiting respects its usage policy; failure degrades to a typed caption.
- **Trade-offs:** online-only; addresses fill gradually for big batches.
- **Risks:** availability/rate limits; coordinates + IP exposed to Nominatim.
- **Future work:** org-hosted/commercial geocoder.
- **Status:** Accepted (owner); flagged for stakeholder approval.

ADR-006 — No persistent storage of user content

- **Decision:** keep working state in memory only; persist nothing user-entered.
- **Context:** privacy-first; no accounts; minimise data-at-rest.
- **Options:** persist drafts (localStorage/IndexedDB); in-memory only.

- **Rationale:** persistence adds a data-at-rest/privacy surface for little benefit in a quick capture-and-export flow.
- **Trade-offs:** closing the tab loses the working set; no resume.
- **Risks:** users may expect autosave.
- **Status:** Accepted (owner).

ADR-007 — Static hosting + PWA offline model

- **Decision:** deploy as static files; provide offline via a service worker with a precached shell and a capped tile cache.
- **Context:** field use with poor connectivity; minimal ops.
- **Options:** online-only static site; static + service-worker offline; native app.
- **Rationale:** static + PWA gives installability and offline without app stores or servers, at near-zero operating cost.
- **Trade-offs:** offline only covers previously viewed tiles; cache-version discipline needed for updates.
- **Risks:** stale caches if `CACHE_VERSION` isn't bumped.
- **Status:** Accepted (owner).

20. Implementation at Suncor & future roadmap

20.1 Implementing at Suncor — primary next steps

If GeoTag is adopted at Suncor, two changes take priority before rollout. Both are localised and consistent with decisions already recorded in this document.

1. Host the app on Suncor infrastructure (not Netlify / public hosting). GeoTag is a set of static files (§16.1), so it can be served from any Suncor-managed static host or intranet web server over HTTPS — no servers, runtime, or database to stand up. Hosting it internally keeps the app, and the network requests it makes, inside Suncor's environment and governance, and lets Suncor apply its own TLS, access control, and the recommended security headers / CSP (§16.5). *Prerequisite:* an approved internal HTTPS static-hosting location.

2. Replace the public map tiles with Suncor's SunMaps / SiteView — if viable and permissible. Today the map uses public OSM / Esri / OpenTopoMap tiles (ADR-004), so tile requests (and the device IP) reach external providers. Pointing the map at Suncor's internal mapping services — **SunMaps** or **SiteView** — would keep location data inside Suncor and could provide plant-accurate basemaps. Whether this is feasible depends on:

- whether SunMaps / SiteView exposes a **standard tile endpoint** (XYZ `{z}/{x}/{y}` or WMTS) that Leaflet can consume — if so, this is a small change to `tileLayerFor()` (`index.html`) and the service worker's `TILE_HOSTS` (`sw.js`), keeping Leaflet as the renderer;
- or, if those products require their own mapping SDK, **replacing Leaflet** with that SDK (a larger change);
- **licensing / permission** to use those basemaps in this tool, and **CORS** support so the exported report image stays untainted (§14.1).

Until this is confirmed viable and permitted, the app keeps working with the public tile providers — so it is an enhancement, not a blocker.

Related: reverse geocoding (Nominatim, ADR-005) would likewise be repointed to a Suncor-approved address/asset lookup at the same time, or simply disabled, since captions are editable by hand.

20.2 Further enhancements

Optional, to be scoped with IT / Digital Services:

- **Internal map / geocoding providers** — see §20.1 (SunMaps / SiteView and an internal geocoder) for the Suncor-specific path; the change points are `tileLayerFor()` / the geocode URL and the service worker's `TILE_HOSTS`.
 - **Strict CSP & security headers** at the static host (§16.5).
 - **Automated testing** (unit/E2E) and a formal **WCAG 2.1 AA audit** with focus-trapping and map accessibility improvements.
 - **Batch / combined report export** (e.g. a single multi-photo PDF) building on the existing per-photo compositor.
 - **Configurable branding** (logo/footer) for report images.
 - **Optional structured export** (CSV/JSON of coordinates and captions) for ingestion into asset/inspection systems.
-

21. Summary for reviewers

QUESTION	ANSWER
Does it have a backend?	No — static files only.
Where are photos processed?	Entirely in the user's browser.
Are photos uploaded anywhere?	No.
What leaves the device?	Only coordinates, to fetch map tiles and an address.
What external services are called?	OSM/Esri/OpenTopoMap tiles and OSM Nominatim — four hosts (§13.5).
Any accounts, cookies, tracking, or analytics?	None.
What is stored, and for how long?	Nothing user-entered persists; only a capped cache of static assets + viewed map tiles (§13.2).
Any secrets/API keys?	None required for the default providers.
Licensing of dependencies?	MIT / BSD / OFL — all permissive, all vendored.
How is it hosted?	Any HTTPS static host or intranet web server.
How is it built/tested/deployed?	No build; manual testing today; deploy = publish files + bump cache version.
Accessibility target?	WCAG 2.1 AA baseline; formal audit outstanding.
Offline?	Yes — app shell and viewed tiles; only fresh tiles + geocoding need the network.
Biggest decisions to approve?	Map/geocode provider, report branding, accessibility certification (§18.3). At Suncor: internal hosting location and SunMaps/SiteView viability (§20.1).
How was it built?	Human-directed, AI-assisted (Claude Code) iterative development — see §17.